

CS439 Project Proposal - Smart Application-Level Scheduling for Wireless Interfaces (WiFi and Bluetooth/ZigBee)

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Project Description

Portable devices such as wireless headphones, mice, and IoT sensors rely on WiFi or Bluetooth/ZigBee for communication. WiFi offers high throughput but consumes more energy, while Bluetooth/ZigBee provide lower energy usage but at reduced throughput. Current systems often stick to one communication interface/protocol and do not intelligently switch between them. However, works such as [2] prove that there is potential for improved performance when using more than one interface/protocol at a time.

This project proposes to recreate and expand on an **application-level scheduling algorithm** that dynamically switches between WiFi and Bluetooth/ZigBee depending on observed data stream characteristics. The algorithm will monitor outgoing data telemetry – such as packet size, urgency, and type – then select the most suitable interface to balance throughput and power efficiency. We will initially aim to create a similar setup and measurement system to [1], and then expand with other metrics and compare their performance.

The goal of this project is to maximize throughput while minimizing energy consumption and improving the overall user experience for portable devices such as cellular phones, wireless mice, headphones, and IoT nodes. High power consumption and degraded user experience – particularly due to persistent network connectivity requirements and mobility-related challenges – have long been recognized issues. Prior work such as [3], [4], and [5] has explored these problems and proposed various solutions. Building on these efforts, we believe this project will contribute meaningfully to addressing these challenges.

Activities

- Literature review on cross-technology communication and interface scheduling.
- Design scheduling policies (heuristic-based or lightweight ML-based).
- Set up hardware with Feather boards and XBee modules.
- Develop monitoring software to capture traffic characteristics.
- Implement scheduling logic at the application layer.
- Test on use cases: audio streaming, mouse input, sensor data.
- Evaluate power consumption vs throughput.

Expected Results

- A prototype demonstrating reading, manipulating, and seamless switching between WiFi and Bluetooth/ZigBee traffic.
- Quantitative results showing power savings without noticeable performance loss.
- Comparison with single-technology baselines.

Equipment and Resources

- $4 \times$ Adafruit Feather boards

- 4 × XBee modules
- 2 × XBee Adapter
- 2 × Breadboards
- 10 × Jumper cables
- Power measurement tools (multimeter, USB power monitor, or profiling software)

Project Plan

- Weeks 1–2: Literature/code/schematic review
- Week 3: Hardware setup (Feather + XBee)
- Week 4: Develop traffic monitoring/reading software
- **By Checkpoint 1:** Be able to read and manipulate WiFi/BLE/ZigBee data, measure power consumption
- Weeks 5–6: Implement switching and scheduling logic
- Week 7: Write test cases + optimize algorithm
- Week 8: Collect performance and power data
- **By Checkpoint 2:** Have scheduling algorithm + test cases implemented
- Week 9: Finalize data collection and generate report and presentation

Division of Work

- Initial research and hardware setup: Both
- Wi-Fi and BLE/ZigBee specifics/testing: split this up down the line
- Power Management and Measurement: Siddharth
- Algorithm Development and Testing: Both
- Data Analysis: Yahav
- Report/Presentation: Both

References

- [1] G. Ananthanarayanan and I. Stoica. Blue-Fi: Enhancing Wi-Fi Performance Using Bluetooth Signals. MobiSys 2009.
- [2] K. Chebrolu and A. Rao. Bandwidth aggregation for real-time applications in heterogeneous wireless networks. IEEE Transactions on Mobile Computing, 2006.
- [3] A. Rahmati and L. Zhong. Context-for-wireless: Context-sensitive energy-efficient wireless data transfer. MobiSys 2007.
- [4] M. Kassar, B. Kervella, and G. Pujolle. An overview of vertical handover decision strategies in heterogeneous wireless networks. Computer Communications, 2008.
- [5] S. Yin, S. Zhang, and Q. Li. Energy-efficient wireless communication for IoT: A survey. IEEE Communications Surveys & Tutorials, 2020.